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Joseph Fernandez, vice president of the Engineered Products Division of Curled Metal Inc. (CMI), and Rajiv Sanwal, group manager of the Mechanical Products Group, had spent the entire day (January 2, 2008) reviewing a new product CMI was about to introduce. (See **Exhibit 1** for organization charts.) Fernandez pondered all that had been said and, turning to Sanwal, commented:

Curled metal cushion pads seem to have more potential than any product we've ever introduced. A successful market introduction could double the sales of this company, as well as compensate for the decline of some existing lines. It almost looks too good to be true.

Sanwal responded, "The people at Kendrick Company are pressing us to sell to them. Since their original test, they've been anxious to buy more. I promised to contact them by the end of the week." "Fair enough," Fernandez said, "but talk to me before you call them. The way we price this could have a significant impact on everything else we do."

The Company

CMI had grown from \$750,000 in sales in 1991 to over \$55 million by 2007. (**Exhibit 2** shows CMI's income statement.) It originally custom-fabricated components for chemical process filtration and other highly technical applications. Company strategy soon evolved from selling metal as a finished product to selling products that used certain types of metal as a raw material.

A big boost came when, in responding to stricter environmental regulations, U.S. auto manufacturers required a high-temperature seal to prevent the escape of very hot exhaust gases. CMI developed a product that it called *Slip-Seal*. Because it could meet the demanding specifications of the automakers, the product captured a large percentage of the available business and CMI had grown rapidly over the past decade. Company management was not confident about maintaining its 80% market share, however, and moved to diversify away from its heavy reliance on Slip-Seal and the auto industry. Thus, when a sales representative from Houston approached CMI's managers with a new application for curled metal technology, management examined it closely.

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The Product

The product discussed by Fernandez and Sanwal was a cushion pad—a key part of the process for driving piles. Piles were heavy beams of wood, concrete, steel, or composite material pushed into the ground as support for a building, bridge, or other structure. They were necessary where ground could shift under the weight of an unsupported structure. Pile driving was typically done with a large crane, to which a diesel or steam hammer inside a set of leads was attached. The leads were suspended over the pile for direction and support. The hammer drove the pile from the top of the leads to a certain depth in the ground (see **Exhibit 3**).

The cushion pads prevented the shock of the hammer from damaging hammer or pile. They sat in a circular “helmet” placed over the top of the pile and were stacked to keep air from coming between striker plate and ram, as shown in **Exhibit 3**. Of equal importance, the pads transmitted energy from the hammer to the pile. A good cushion pad had to transmit force without creating heat, and still remain resilient enough to prevent shock. With an ineffective pad, energy transmitted from the hammer would be given off as heat, and the pile could start to vibrate and possibly crack.

Despite their importance to the pile-driving process, little attention had been paid to pads by most of the industry. Originally, hardwood blocks were used. Their cushioning was adequate, but availability was a problem and performance was poor. Constant pounding quickly destroyed the wood’s resiliency, heat built up, the wood often ignited, and blocks had to be replaced frequently.

Most of the industry eventually shifted to stacks of alternate layers of ½-inch-thick aluminum plate and 1-inch-thick micarta slabs. (These were not fabricated, but simply pieces of aluminum and micarta cut to specific dimensions.) By 2008, micarta slabs, and some phenolic plastic pads, were the conventional pads most often used and had cost and performance characteristics similar to each other. Current pads came in a variety of standard diameters, the most common being 11½ inches. Diameter was determined by the size of the helmet, which varied with the size of the pile.

CMI Cushion Pad

Curled metal was continuous metal wire that was flattened and then wound into tight, continuous ringlets. This allowed the metal to stretch in length and width and gave it three-dimensional resiliency. Because it could be made of various metals (e.g., copper, Monel, and stainless steel), curled metal could be made to withstand almost any temperature or chemical. Stacking many layers could produce a shock mount, an airflow corrector, or a highly efficient filter. Tightly compressed curled metal produced the Slip-Seal for auto applications or, when calendered (i.e., curled metal ringlets were compressed between rollers to make a smooth, tight band) and wound around an axis, a cushion pad for pile driving.

CMI purchased wire from outside vendors and performed the flattening and curling operations in-house. The CMI pad started with curled metal calendered to about 1 inch thick and wound tightly around the center of a flat, metallic disk until the desired diameter was reached. A similar disk was placed on top, with soldered tabs folded down to hold it all together. The pad was then coated with polyvinyl chloride to enhance its appearance and disguise the contents (see **Exhibit 4**).¹ This manufacturing process allowed any diameter pad, from the standard minimum of 11½ inches to over 30 inches for a custom application, to be produced from the same band of curled metal.

¹ Managers at CMI were concerned that other manufacturers might discover this new application for curled metal and enter the business before CMI could get patent protection. CMI had a number of competitors; most were substantially smaller than CMI and none, thus far, had shown a strong interest or competence in technical, market, or product development.

Comparative Performance

After struggling to find a responsible contractor to use the product and monitor its performance, CMI persuaded Kendrick Foundation Company of Baltimore, Maryland, to try its pads on a papermill expansion in Newark, Delaware. The job required 300 55-foot piles driven 50 feet into the ground. The piles were 10-inch and 14-inch steel H-beams; both used an 11½-inch helmet and, thus, 11½-inch cushion pads. The total contractor revenue from the job was \$225,000 (\$15 per foot of pile driven).

Kendrick drove a number of piles using conventional ¼-inch-thick cushion pads to determine their characteristics for the job. Eighteen were placed in the helmet and driven until they lost resiliency. Pads were added, and driving continued until a complete set of 24 was sitting in the helmet. After these were spent, the entire set was removed and the cycle repeated. The rest of the job used the CMI pads. Four were initially installed and driven until 46 piles had been placed. One pad was added and the driving continued for 184 more piles. Another pad was placed in the helmet, and the job was completed. Comparable performances for the entire job were extrapolated as follows:

	Conventional Pads	CMI Pads	Economic value, to calculate.
1. Feet driven per hour while pile driver was at work (does not consider downtime)	150	200	- 33% more efficient
2. Piles driven per set of pads	15	300	- 2000% more piles
3. Number of pads per set	24	6	- 75% less pads
4. Number of sets required	20	1	- 95% less sets needed
5. Number of set changes	20	1	- 95% less sets changed
6. Time required for change per set	20 minutes	4 minutes	- 80% less time needed
7. Kendrick cost per set	\$150	Not charged	- determine the price

Although the CMI pads drove piles 33% faster and lasted for the entire job, Sanwal felt these results were unusual. He believed that a curled metal set life of 10 times more than conventional pads, and a performance increase of 20%, were probably more reasonable, because he was uncertain that CMI pads in larger sizes would perform as well.

WTP is indicated here?

Industry Practice

Industry sources indicated that about 75% of pile-driving contractors owned their hammers, and most owned at least one crane and set of leads. To determine contractors' costs, CMI studied expenses of smaller contractors who rented equipment for pile-driving jobs. These data were available and avoided the problem of allocating the cost of a purchased crane or hammer to a particular job.

Standard industry practice for equipment rental used a three-week month and a three-day workweek.² This was simply tradition, but most equipment renters set their rates this way. The cost of renting the necessary equipment and the labor cost for a job similar to that performed by Kendrick were estimated as shown in **Table A**.

² This means that a contractor who rented equipment for one calendar month was charged only the "three-week" price, but had the equipment for the calendar month. The same was true of a "three-day week." Contractors tried to use equipment for as much time per week or per month as possible; they rented it on a "three-week" month but used it on a "4.33-week" month.

Table A Equipment Rental, Labor, and Overhead Costs

	Per Standard		Per Hour	Average Cost per Real Hour ^a
	Month	Week		
1. Diesel hammer	\$13,500–21,600	\$4,500–7,200	\$187.50–300.00	\$102
2. Crane	24,000–30,000	8,001–10,002	333.00–420.00	156
3. Leads @ \$60 per foot per month (assume 70 feet)	4,200	1,401	58.32	24
4. Labor ^b —3 laborers @ \$18–24/hour each			54.00–72.00	63
1 crane operator			24.00–36.00	30
1 foreman			36.00–42.00	39
5. Overhead ^c (office, trucks, oil/gas, tools, etc.)			300.00	300

(Casewriter's note: Please use average cost per real hour in all calculations, for uniformity in class discussion.)

^a These costs were calculated from a rounded midpoint of the estimates. Hammer, crane, and lead costs were obtained by dividing standard monthly costs by 4.33 weeks per month and 40 hours per week.

^b Labor was paid on a 40-hour week and 4.33-week month. One-shift operation (40 hours/week) was standard in the industry.

^c Most contractors calculated overhead on the basis of "working" hours, not standard hours.

Hidden costs also played a role. For every hour actually spent driving piles, a contractor could spend 20 to 40 minutes moving the crane into position. Another 10% to 15% was added to cover scheduling delays, mistakes, and other unavoidable problems. Thus, the real cost per hour was usually substantially more than the initial figures showed. Reducing the driving time or pad changing time did not usually affect the time lost on delays and moving.

All of these figures were based on a job that utilized 55-foot piles and 11½-inch pads. Although this was a common size, much larger jobs requiring bigger material were frequent. A stack of 11½-inch micarta pads weighed between 30 and 40 pounds; the 30-inch size could weigh seven to eight times more. Each 11½-inch CMI pad weighed 15½ pounds. Bigger sizes, being more difficult to handle, could contribute significantly to unproductive time on a job. (See **Exhibit 5**.)

Most contracts were awarded on a revenue-per-foot basis. Contractors bid by estimating the time it would take to drive the specified piles the distance required by the architectural engineers. After totaling costs and adding a percentage for profit, they submitted figures broken down into dollars per foot. The cost depended on the size of the piles and the type of soil to be penetrated. The \$15 per foot that Kendrick charged was not atypical, but prices could be considerably greater.

very low 'non-price acquisition cost' to Kendrick for using CMI.

More Test Results

CMI's management was very pleased by how its cushion pads had performed for Kendrick. They lasted the entire job, eliminating downtime required for changeover, and other advantages became apparent. For example, after 500 feet of driving, the average temperature for the conventional pads was 600° to 700°F, which created great difficulty when they had to be replaced. The crew handling them was endangered and much time was wasted waiting for pads to cool, accounting for a major portion of time lost to changeovers. CMI pads, in contrast, never went above 250°F and were handled almost immediately with protective gloves. Thus, substantial energy lost in heat by the other pads was being used more efficiently to drive the piles with CMI pads. Also, the outstanding resiliency of CMI's product seemed to account for a 33% faster driving time, which translated into significant savings.

In talking with construction site personnel, CMI also found that most were becoming wary of conventional pads' potential dangers, due to the heat generated and materials used. Many expressed a desire to use other material and were pleased that CMI pads contained no hazardous materials. And while CMI was happy with the results, Kendrick was ecstatic: it wanted to buy more pads and pressed Sanwal to quote prices.

To confirm the Kendrick test results, therefore, CMI asked Corey Construction to try the pads on a job in Pennsylvania that required 300 45-foot concrete piles to be driven 40 feet into the ground. Conventional pads of 11½ inches were again the comparison. Total job revenue was \$324,000 or \$27 per foot. Corey paid \$120 for each set of 12 micarta pads used. Results were as follows:

Corey likely has higher WTP for CMI pads than Kendrick due to higher revenue per foot.

	Conventional Pads	CMI Pads
1. Feet driven per hour while pile driver was at work (does not consider downtime)	160	200
2. Piles driven per set of pads	6	300
3. Number of pads per set	12	5
4. Number of sets required	50	1
5. Number of set changes	50	1
6. Time required for change per set	20 minutes	4 minutes
7. Corey cost per set	\$120	Not charged

The Market

There were virtually no statistics from which a potential U.S. market size for cushion pads could be determined, so Sanwal made several assumptions based on information he could gather. A report by *Construction Engineering* magazine estimated that about 13,000 pile hammers were owned by companies directly involved in pile driving. Industry sources estimated that another 6,500 to 13,000 hammers were leased. Sanwal assumed that this total of 19,500 to 26,000 hammers would operate about 25 weeks per year (because of seasonality) and be used 30 hours per week (because of moving time, repairs, scheduling problems, and other factors).

Sanwal also assumed that an average actual driving figure (including time to change pads and so on) for most jobs was 20 feet per hour, which amounted to between 290 and 390 million feet of piles driven annually. To be conservative, he also assumed that a set of curled metal pads (four initially installed, plus two added after the originals lost some resiliency) would drive 10,000 feet.

Purchasing Influences

In the pile-driving business, as in other parts of the construction industry, a number of entities participated in purchases. CMI management was able to identify six types of influences:

1. *Pile hammer manufacturers.* Some manufacturers sold hammers in the United States and many were imported from Europe and Japan. The leading domestic producer in 2008 was Prometheus Iron Works, whose Model #1 was the standard used by architectural engineers specifying equipment for a job. Sanwal did not feel these manufacturers would purchase a large dollar volume of cushion pads, but they could influence recommendations.
2. *Architectural/consulting engineers.* Pile driving required much expertise in determining the needs of a construction project, including thorough stress analysis and other mathematical analysis. Because of the risks in building the expensive projects supported by piles, the industry looked to architectural/consulting engineers as the ultimate authorities on all aspects of the business. These firms were very detailed in specifying the materials and techniques for a project. They always specified hammers and frequently mentioned pads. CMI felt that, while no sales would come from these people, they could be a key influence on purchasing decisions.
3. *Soil consultants.* These consultants were similar to the architectural/consulting engineers, but were consulted only when extraordinary conditions existed.
4. *Pile hammer distributing/renting companies.* This group provided pads to the contractors. In fact, renting companies often included the first set of pads free. CMI management felt that these companies would handle the cushion pads they could most easily sell and might even hesitate to provide pads that enabled a contractor to return equipment faster.
5. *Engineering/construction contractors.* The contracting portion of the industry was divided among large international firms and smaller independents. The former typically participated in the bigger, more sophisticated jobs. Companies like Bechtel and Conmaco contracted to drive piles, designed jobs, specified material, and even manufactured their own equipment. Sanwal believed that, to get CMI pads used on bigger, complex construction projects, CMI would have to solicit this group actively on a very sophisticated level.
6. *Independent pile-driving contractors.* These contractors represented the “frontline buying influence.” Their primary objective was to make money. They were very knowledgeable about the practical aspects of pile driving, but not very managerially sophisticated.

No national industry associations influenced this business, but some regional organizations played a minor part. Contractors and others talked freely, although few were willing to reveal competitive secrets. CMI was unsure how important word-of-mouth communication could be. Little was published about the pile-driving industry, although magazines like *Oklahoma Contractor* occasionally reported on pile-driving jobs and featured advertising by suppliers to the trade, mostly equipment dealers and supply houses. One supplier, Amalgamated Pile and Fitting Corporation, sponsored professional-level “Piletalk” seminars in various cities, bringing designers, contractors, and equipment developers together “to discuss practical aspects of installation of driven piles.”

Another potential influence was Professor Stephen McCormack of Pennsylvania A&M University. He had established a department to study pile driving and was a respected authority on its theoretical aspects. Sophisticated engineering/construction firms and many architectural consultants were familiar with his work and helped support it. CMI felt that his endorsement of the operational

performance of CMI pads would greatly enhance industry acceptance. The company submitted its pads for testing by Dr. McCormack in the fall of 2007, and although the final results were not yet available he had expressed considerable enthusiasm. Final results were expected by early 2008.

Competitive Products and Prices

The pile-driving industry had paid very little attention to cushion pads. Everyone used them and took them for granted, but no one attempted to promote pads. No manufacturers dominated the business. In fact, most pads came unbranded, having been cut by small, anonymous job shops.

Distribution of pads was also ambiguous. Hammer sales and rental outlets provided them, heavy construction supply houses carried them, pile manufacturers sometimes offered them, and an assortment of other outlets occasionally sold them.³ The smaller conventional pads sold for \$6 to \$9 each; larger ones sold for between \$15 and \$30. Nine dollars each was typical for 11½-inch pads. The profit margin for a distributor was usually adequate—in the area of 30% to 40%—but the dollar profit did not compare well with that of other equipment lines the distributors carried. Most outlets carried pads as a necessary part of the business, but none featured them as a work-saving tool.

CMI management felt it could be totally flexible in establishing a strategy and organization to approach the market. It considered a direct sales force and its own distribution outlets, but eventually began to settle on signing construction-oriented manufacturers' representatives,⁴ who would sell to a variety of distributors and supply houses. CMI feared an uphill struggle to convince the industry's sales and distribution channels that there really was a market for the new pad. CMI expected difficulty in finding outlets willing to devote the attention necessary for success, but it also felt that once the initial barriers were penetrated, most of the market would be anxious to handle the product.

Decisions

Sanwal had projected cost data developed by his manufacturing engineers. **Exhibit 6** shows two sets of numbers: one utilized existing equipment; the other reflected the purchase of \$150,000 of permanent tooling. In both cases, the estimated volume was 250 cushion pads per month. Additional equipment could be added at a cost of \$225,000 per 250 pads per month of capacity, including permanent tooling that could be purchased for \$150,000.

Both sets of numbers assumed the manufacture of only one pad size; in other words, the numbers in the 11½-inch size were based on manufacturing only that size for a year. This was done because CMI had no idea of the potential sales mix among product sizes. Management knew that 11½ inches was the most popular size, but the information available on popularity of the other sizes was vague. CMI finance personnel believed these numbers would not vary dramatically with a mix of sizes.

Corporate management usually burdened CMI products with a charge equal to 360% of direct labor to cover the overhead of its large engineering staff. Sanwal was uncertain how this would apply to the new product, because little engineering had been done and excess capacity was to be used initially for manufacturing. Although it was allocated on a variable basis, he thought he might

³ Supply houses were "hardware stores" for contractors and carried many products, including lubricants, work gloves, and maintenance supplies. Distributors, in contrast, were more equipment oriented and sold a narrower line of merchandise.

⁴ Manufacturers' representatives were agents (sometimes individuals, sometimes organizations) who sold noncompeting products for commission. They typically did not take title to the merchandise and did not extend credit.

Fixed/sunk cost
should not be used
in pricing

consider the overhead “fixed” for his analysis. Corporate management also expected a contribution margin after all manufacturing costs of 40% to 50% of selling price.

Sanwal was enthusiastic about the potential of this new product. The Engineered Products Division was particularly pleased to offer something with such high dollar potential, especially since a “large” customer for this division typically purchased only about \$30,000 per year. But Sanwal was uncertain how to market the pads and other aspects of business strategy for the product as well as the potential implications for CMI activities ranging from capital investment strategy thru production and distribution. Some had suggested advertising and promotion, for instance, but there were no ad or promotional precedents for this product.

For the moment, however, the primary consideration was pricing. Sanwal had promised to call Kendrick Foundation Company by the end of the week, and Fernandez was anxious to review the decision with him. He hoped other prospects would be calling as soon as word about the pads’ test performance got around.

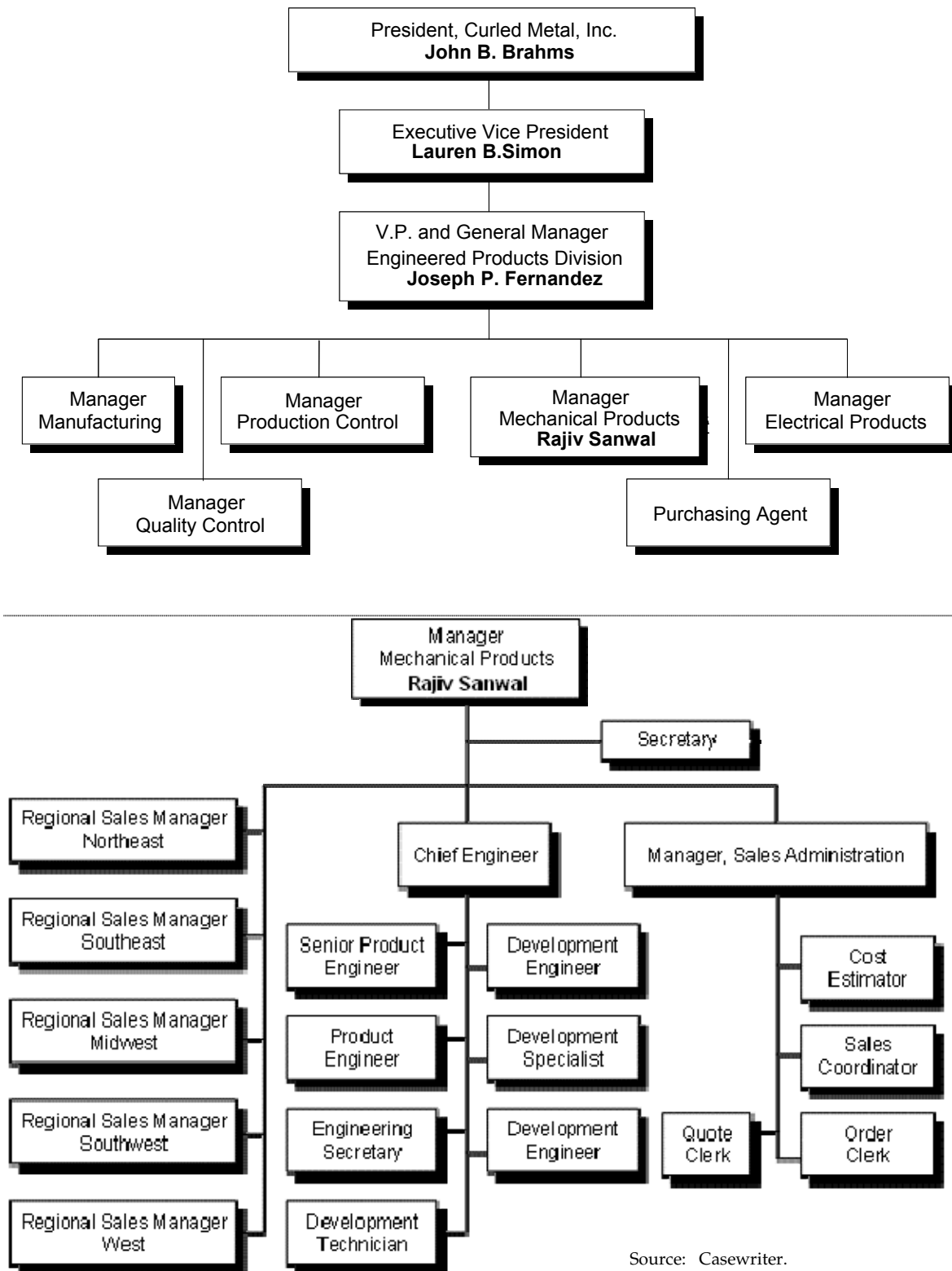
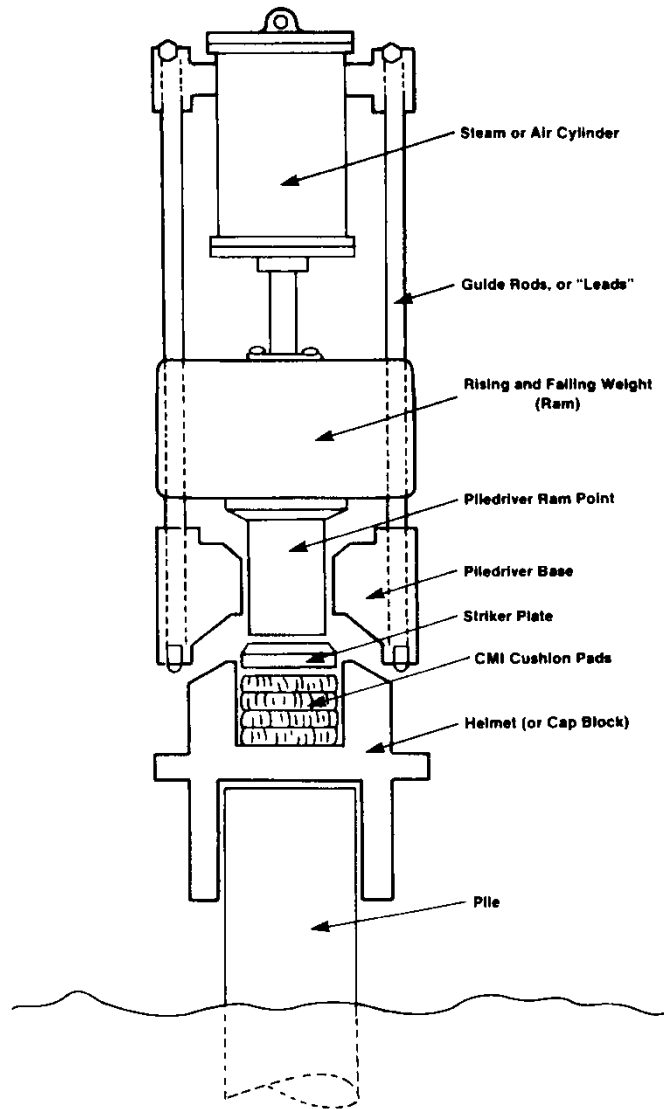
Exhibit 1 Engineered Products Division Organization Chart

Exhibit 2 CMI's Income Statement

December 31	2007	2006
Net sales	\$55,573,284	\$61,395,171
Costs and expenses		
Cost of sales	33,764,781	35,279,043
Selling expenses	8,929,188	8,133,960
General and administrative expenses	<u>6,612,873</u>	<u>7,087,584</u>
	49,306,842	50,500,587
Income from operations	6,266,442	10,894,584
Other income (expense)		
Dividend income	626,856	—
Interest income	218,898	559,833
Interest expense	<u>(121,908)</u>	<u>(94,128)</u>
	723,846	465,705
Income before income taxes	6,990,288	11,360,289
Provision for income taxes	<u>3,506,490</u>	<u>5,679,846</u>
Net income	<u><u>3,483,798</u></u>	<u><u>5,680,443</u></u>
Net income per share	\$4.17	\$6.48

Exhibit 3 Typical Steam- or Air-Operated Pile Driver with Helmet and Cushion Pad

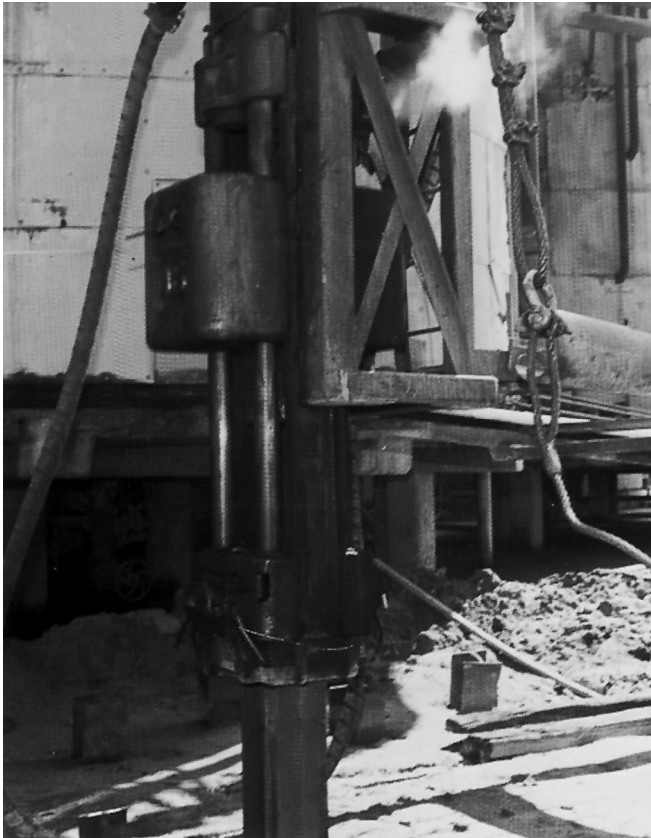


A schematic diagram of typical pile driver



Pile hammer inside leads driving a steel H-beam into the ground

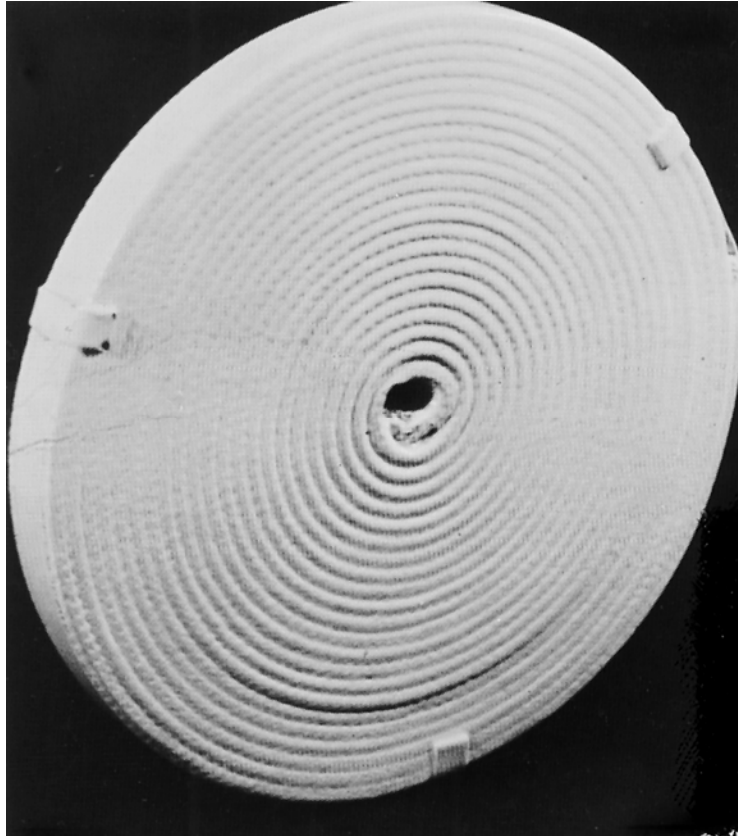
Exhibit 3 (continued)



Close-up of hammer driving pile (most of the pile is already in the ground)



CMI pile-driving pad in position in helmet

Exhibit 4 Close-Up of CMI Curled Metal Cushion Pad for Pile Driving

The calendered curled metal is wound tightly around the central point of a flat metallic disk. (The disk is on the back side of the pad from this view.) Soldered tabs secure the curled metal to the disk. The entire structure is coated with polyvinyl-chloride.

Exhibit 5 Curled Metal Cushion Pad Standard Sizes

Diameter (inches)	Thickness (inches)	Weight (pounds)
11½	1	15½
14	1	23
17½	1	36
19¾	1	48
23	1	64
30	1	110

Exhibit 6 Two Sets of Projected Manufacturing Costs

	Size					
	11 ½"	14"	17 ½"	19 ¾"	23"	30"
Estimates per Pad with Existing Equipment						
<i>Variable</i>						
Material	\$46.92	\$61.71	\$95.43	\$121.17	\$159.48	\$287.07
Labor	<u>86.40</u>	<u>99.21</u>	<u>150.06</u>	<u>171.21</u>	<u>207.48</u>	<u>355.08</u>
Total variable	133.32	160.92	245.49	292.38	366.96	642.15
Fixed factory overhead						
@ 360% direct labor	<u>311.04</u>	<u>357.15</u>	<u>540.21</u>	<u>616.35</u>	<u>746.94</u>	<u>1,278.30</u>
Total manufacturing cost	\$444.36	\$518.07	\$785.70	\$908.73	\$1,113.90	\$1,920.45
Estimated with Purchase of \$150,000 of Permanent Tooling						
<i>Variable</i>						
Material	\$46.92	\$61.71	\$95.43	\$121.17	\$159.48	\$287.07
Labor	<u>34.92</u>	<u>45.75</u>	<u>65.55</u>	<u>80.85</u>	<u>91.71</u>	<u>168.27</u>
Total variable	81.84	107.46	160.98	202.02	251.19	455.34
Fixed factory overhead						
@ 360% direct labor	<u>125.70</u>	<u>164.70</u>	<u>235.98</u>	<u>291.06</u>	<u>330.15</u>	<u>605.76</u>
Total manufacturing cost	\$207.54	\$272.16	\$396.96	\$493.08	\$581.34	\$1,061.10

Note: Estimated volume was 250 cushion pads per month.